SESSION 7 · MINI-ASSESSMENT REPORT

# Career Guidance for 10th and 12th Grade Students

*Design and Implementation of a Multi-Step Prompt Workflow*

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# 1. Introduction

Large language models can carry out complex tasks when guided by carefully designed prompts. Asking a single prompt to perform several tasks at once, however, often produces inconsistent formatting, missed details, or unreliable results.

Prompt chaining addresses this problem by breaking a larger task into smaller, specialised stages. Each stage performs one specific function, and the output of one stage becomes the input for the next.

This report presents the design and implementation of a three-step prompt workflow for career guidance triage aimed at students in Grade 10 and Grade 12. The workflow takes an unstructured message from a student describing their academic situation, interests, and concerns, and progressively transforms it into a structured result that could be used within an automated career-counselling support system.

The workflow follows this sequence:

**Raw Student Query → Profile Extraction → Guidance Assessment → Structured JSON Output**

Each stage applies a prompting technique suited to its purpose:

# 2. Objective

This mini-assessment set out to design and test a three-step prompt chain for processing a query from a 10th or 12th grade student seeking career guidance. The specific objectives were to:

- Extract the important academic and personal information from an unstructured student message using few-shot prompting.
- Analyse the extracted information and determine the most appropriate guidance category using structured reasoning.
- Convert the results into a valid JSON object that could be processed by another software system.
- Demonstrate that the output of each stage is used as the input for the next.

# 3. Workflow Design

The workflow was designed as a sequential three-stage pipeline, illustrated in Figure 1.

**[ Raw Student Query ]**
↓ Step 1 · Few-Shot Prompting
**[ Extracted Student Profile ]**
↓ Step 2 · Chain-of-Thought Reasoning
**[ Guidance Category / Recommendation ]**
↓ Step 3 · JSON Formatting
**[ Structured JSON Output ]**

*Figure 1: Prompt Chaining Workflow*

Separating the workflow into three stages ensures that each prompt has a single, focused responsibility. Step 1 identifies the information relevant to the student's situation; Step 2 makes a judgement based on that information; and Step 3 packages the result for downstream use.

This separation keeps each individual prompt simple and makes the workflow easier to inspect, since every intermediate result can be reviewed on its own.

# 4. Methodology

The workflow was implemented using an AI assistant interface (Claude). The following student query was used as the input for the experiment:

**Student Query**

```
"I'm in 12th grade, PCM group. I'm good at maths and physics and my marks are
strong, but honestly what I enjoy most is graphic design — I spend most of my
free time doing digital art and learning design software. My parents want me
to prepare for JEE and become an engineer, and I haven't told them how I
really feel. My board exams are in three months and I don't know whether to
keep pushing through JEE prep or start looking into design as an option."
```

## 4.1 Step 1: Profile Extraction Using Few-Shot Prompting

**Purpose**

The purpose of the first stage was to extract the most important academic and personal information from the student's unstructured message. Few-shot prompting was chosen because worked examples demonstrate the relationship between an input and its desired output, while also establishing a consistent format for the extracted information. Three examples were provided before the actual student query.

**Prompt Used**

```
I'm building a simple career-guidance support workflow for school students.
For the first step, I need you to read a student's message and pull out the
details that would be useful for deciding what kind of guidance they need.

Use the examples below as a guide for what to look for and how to format
your answer.

Example 1:
Student message: "I'm in 10th grade. I really like biology and enjoy watching
nature documentaries, but I'm also decent at maths. I haven't thought much
about which stream to choose yet."
Output:
Grade level: 10th
Favourite subjects: biology, maths
Interests/hobbies: nature documentaries
Career direction so far: undecided
Family/external pressure: none mentioned
Emotional tone: relaxed, low urgency

Example 2:
Student message: "I'm in 12th grade, Commerce stream. I've always wanted to
become a chartered accountant like my father, and I'm currently preparing
for the CA Foundation exam alongside my board exams. It's a lot of work but
I feel motivated."
Output:
Grade level: 12th (Commerce)
Favourite subjects: accountancy, business studies (implied)
Interests/hobbies: none specifically mentioned
```

```
Career direction so far: Chartered Accountancy (CA Foundation in progress)
Family/external pressure: following father's profession
Emotional tone: motivated, some workload stress but manageable

Example 3:
Student message: "I'm in 12th grade, PCB group, and I scored well in my
practicals, but I honestly can't stand the sight of blood and don't want to
do medicine. My parents keep telling relatives I'll become a doctor and I
don't know how to correct them."
Output:
Grade level: 12th (PCB)
Favourite subjects: unclear (practicals strong)
Interests/hobbies: not mentioned
Career direction so far: none decided; actively against medicine
Family/external pressure: parents have publicly assumed medicine
Emotional tone: uncomfortable, avoidant, seeking correction

Now analyse this student query using the same style:
"I'm in 12th grade, PCM group. I'm good at maths and physics and my marks
are strong, but honestly what I enjoy most is graphic design — I spend most
of my free time doing digital art and learning design software. My parents
want me to prepare for JEE and become an engineer, and I haven't told them
how I really feel. My board exams are in three months and I don't know
whether to keep pushing through JEE prep or start looking into design as
an option."
```

**Output Generated**

```
Grade level: 12th (PCM)
Favourite subjects / academic strength: mathematics and physics (strong marks)
Interests/hobbies: graphic design, digital art, design software
Career direction so far: uncertain — weighing engineering (JEE) against design
Family/external pressure: parents expect engineering via JEE; true feelings
not yet disclosed
Timeline: board exams in approximately three months; JEE preparation ongoing
Emotional tone: confused, hesitant, unspoken tension with parents
```

**Discussion**

The first step successfully converted an unstructured student message into a concise list of relevant information, identifying several factors that matter for career-guidance triage:

- The student's grade level and academic stream.
- Whether academic strength and personal interest point in the same direction.
- Any career direction the student has already considered or is being pushed toward.
- Family or external pressure that may be shaping the decision.
- The student's emotional tone and any relevant time pressure, such as upcoming exams.

This output was then used directly as the input for the second stage.

## 4.2 Step 2: Guidance Assessment Using Chain-of-Thought Reasoning

**Purpose**

The purpose of the second stage was to determine the most appropriate guidance category for the student. Unlike profile extraction, judging the right category requires weighing several factors together, so the workflow used a structured reasoning approach that considers whether interest and aptitude align, how much external or family pressure is involved, how time-sensitive the decision is, and whether the core issue is a lack of information or an unresolved conflict.

**Prompt Used**

```
Now that the important details have been extracted, the next step is to
decide what kind of guidance this student needs.

Look at the information below and think through the decision carefully.
Consider whether the student's interests and academic strengths point in
the same direction, how much family or external pressure is involved, how
time-sensitive the situation is, and whether the core issue is a lack of
information or an unresolved conflict between the student and others.

Use the following guide:
On-track: interest, aptitude, and stated goal are aligned; continue current
preparation with periodic check-ins.
Exploratory: interests or strengths are not yet clear; recommend exposure
activities (aptitude tests, mentorship, taster sessions) with no urgent
decision needed.
Conflict – needs dialogue: a clear mismatch exists between the student's
genuine interest and the path being pursued, particularly where family
expectations are involved; requires a facilitated conversation and
exploration of hybrid paths.
Time-critical decision: a hard deadline (stream choice, application,
entrance-exam registration) is imminent with no resolution reached,
requiring immediate structured decision support.

Explain your decision in a few numbered points. Base your answer only on
the information provided below.

Extracted student profile:
Grade level: 12th (PCM)
Favourite subjects / academic strength: mathematics and physics (strong marks)
Interests/hobbies: graphic design, digital art, design software
Career direction so far: uncertain – weighing engineering (JEE) against design
Family/external pressure: parents expect engineering via JEE; true feelings
not yet disclosed
Timeline: board exams in approximately three months; JEE preparation ongoing
```

```
Emotional tone: confused, hesitant, unspoken tension with parents
```

**Output Generated**

```
1. The student's demonstrated aptitude (strong maths and physics marks) does
   not match their stated enthusiasm (graphic design). This is an interest-
   aptitude divergence, not a lack of ability in either area.
2. The current path (JEE/engineering) is primarily driven by parental
   expectation rather than the student's own preference, and the student
   has not yet communicated their real interest — this is an unresolved
   conflict, not a resolved career fit.
3. There is moderate time pressure (board exams in three months), but this
   is not yet an irreversible decision point: portfolio-based design routes
   and hybrid fields such as product design, UX/UI, or engineering-plus-
   design programmes can often be explored alongside board preparation.
4. Because the core issue is a communication and values conflict rather than
   a lack of career information, the priority is facilitated dialogue with
   family and exploration of hybrid paths, rather than an immediate switch
   of exam registration.

Guidance category: Conflict – needs dialogue (secondary flag: time-sensitive,
given the three-month exam window)
```

**Discussion**

The query was classified under the “Conflict – needs dialogue” category because several factors coincided: a clear divergence between academic aptitude and genuine interest, an external expectation that has not been discussed openly with the student's parents, and a moderate but not immediate time constraint.

This reasoning stage worked from the information extracted in Step 1 rather than analysing the original student query independently, demonstrating the chaining relationship between the two prompts. The final guidance category was then passed to the third stage.

## 4.3 Step 3: Structured Output Using JSON

**Purpose**

The purpose of the final stage was to convert the results of the previous two stages into a structured format. JSON was chosen because it is consistently structured and easy for software systems to process — a result such as this could potentially be consumed by a school counselling dashboard, a student-support ticketing system, or a parent-teacher communication tool.

**Prompt Used**

```
Convert the results from the previous two stages of this career-guidance
workflow into a valid JSON object.

Return only the JSON object. Do not include explanations, headings, or
markdown.

Use exactly these three keys:
"student_profile": a list of the important details extracted from the query
```

```
"guidance_category": the final category decided during the previous step
"summary": one sentence explaining the student's situation and why it
requires that guidance category

Results from Step 1:
Grade level: 12th (PCM)
Favourite subjects / academic strength: mathematics and physics (strong marks)
Interests/hobbies: graphic design, digital art, design software
Career direction so far: uncertain – weighing engineering (JEE) against design
Family/external pressure: parents expect engineering via JEE; true feelings
not yet disclosed
Timeline: board exams in approximately three months; JEE preparation ongoing
Emotional tone: confused, hesitant, unspoken tension with parents

Result from Step 2:
Guidance category: Conflict – needs dialogue (time-sensitive)
```

**Final JSON Output Generated**

```
{
  "student_profile": [
    "Grade level: 12th (PCM)",
    "Academic strength: mathematics and physics",
    "Genuine interest: graphic design and digital art",
    "Family expectation: engineering via JEE, not yet discussed openly",
    "Timeline: board exams in approximately three months",
    "Emotional tone: confused, hesitant"
  ],
  "guidance_category": "conflict_needs_dialogue",
  "summary": "The student shows strong aptitude in maths and physics but a
  genuine passion for graphic design that conflicts with undisclosed
  parental expectations of engineering, requiring facilitated family
  dialogue and exploration of hybrid design-engineering pathways before
  board exams in three months."
}
```

## 5. Results and Discussion

The three-stage workflow successfully processed the original student query, carrying it from an unstructured message through to a structured JSON result.

**Table 1: Summary of the Prompt Workflow**

| Step | Task | Prompting Technique | Output |
|---|---|---|---|
| Step 1 | Extract the student's academic and personal profile | Few-shot prompting | Structured profile list |

| | | | |
|---|---|---|---|
| Step 2 | Determine the guidance category / recommended path | Chain-of-thought reasoning | Guidance classification |
| Step 3 | Format the results | JSON output | Machine-readable JSON object |

The workflow demonstrates the advantage of giving each prompt a specific, narrow purpose. The first prompt focuses on extracting information and establishing consistency through worked examples. The second prompt uses that extracted information to make a values- and context-aware judgement about the type of guidance needed. The final prompt performs no further analysis and instead focuses purely on formatting the existing results correctly.

This separation makes the workflow easier to review, since each intermediate result can be inspected on its own. It also means the final output follows a consistent schema containing the three required fields:

- student_profile
- guidance_category
- summary

# 6. Conclusion

This mini-assessment demonstrated the design and implementation of a three-step prompt chaining workflow for career guidance triage aimed at 10th and 12th grade students. The workflow applied three distinct prompt engineering techniques:

- Few-shot prompting was used to extract important academic and personal information from an unstructured student query.
- Chain-of-thought reasoning was used to weigh interest-aptitude alignment, family pressure, timeline urgency, and the underlying nature of the issue to determine the student's guidance category.
- JSON formatting was used to convert the final results into a structured, machine-readable output.

The completed workflow followed the sequence Extract → Reason → Format, with the output of each stage passed forward as the input to the next, forming a complete prompt chain.